Spinor solution and CPT

Peskin & Schroeder : An Introduction to Quantum Field Theory § 3.3~3.4 & 3.6

Mattew D. Schwartz : Quantum Field Theory and the Standard Model Chapter11

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1. Chirality, helicity and spin

In the Weyl basis, the gamma matrices and Lorentz generator $S^{\mu\nu}$ are block diagonal.

$$\gamma^{\mu} = \begin{pmatrix} 0 & \sigma^{\mu} \\ \overline{\sigma}^{\mu} & 0 \end{pmatrix} \qquad S^{\mu\nu} \equiv \frac{i}{4}[\gamma^{\mu}, \gamma^{\nu}]$$

Under infinitesimal Lorentz transformation,

$$\psi \to \psi' = \begin{pmatrix} 1 - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\sigma} - \frac{1}{2}\boldsymbol{\beta}\cdot\boldsymbol{\sigma} & \\ & 1 - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\sigma} + \frac{1}{2}\boldsymbol{\beta}\cdot\boldsymbol{\sigma} \end{pmatrix}\psi$$

Each two-component field transforms independently, and the opposite boost signs distinguish left-and right-handed spinors.

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

The handedness of a spinor is also known as its **chirality**.

1. Chirality, helicity and spin

To project out the left- or right-handed Weyl spinor by using gamma matrices

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$$

Before doing this, let's check the matrix γ^5 properties

$$\gamma^{5\dagger}=\gamma^5$$

$$(\gamma^5)^2=1$$

$$\{\gamma^5,\gamma^\mu\}=0$$

$$[\gamma^5,S^{\mu\nu}]=0$$

1. Chirality, helicity and spin

$$\gamma^5 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$$

Left- and right-handed spinors are eigenstates of γ^5 with eigenvalues ∓ 1.

$$\gamma^5 \psi = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} \psi_{\mathrm{L}} \\ \psi_{\mathrm{R}} \end{pmatrix} = \begin{pmatrix} -\psi_{\mathrm{L}} \\ +\psi_{\mathrm{R}} \end{pmatrix}$$

Then, the projection operators

$$P_L = \frac{1-\gamma^5}{2} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_R = \frac{1+\gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_L P_R = 0$$

$$P_L \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix}, P_{\mathrm{R}} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} 0 \\ \psi_R \end{pmatrix}$$

1. Chirality, helicity and spin

In the Weyl basis, the Dirac equation is

$$\begin{pmatrix} -m & i\sigma \cdot \partial \\ i\bar{\sigma} \cdot \partial & -m \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0$$

$$i\sigma^{\mu}\partial_{\mu}\psi_R = m\psi_L, \qquad i\bar{\sigma}^{\mu}\partial_{\mu}\psi_L = m\psi_R$$

For the Lagrangian density, mass term couples the two chiral components.

$$m\bar{\psi}\psi = m\left(\psi_L^{\dagger}\psi_R + \psi_R^{\dagger}\psi_L\right)$$

So, the mass mixes the left- and right-handed states.

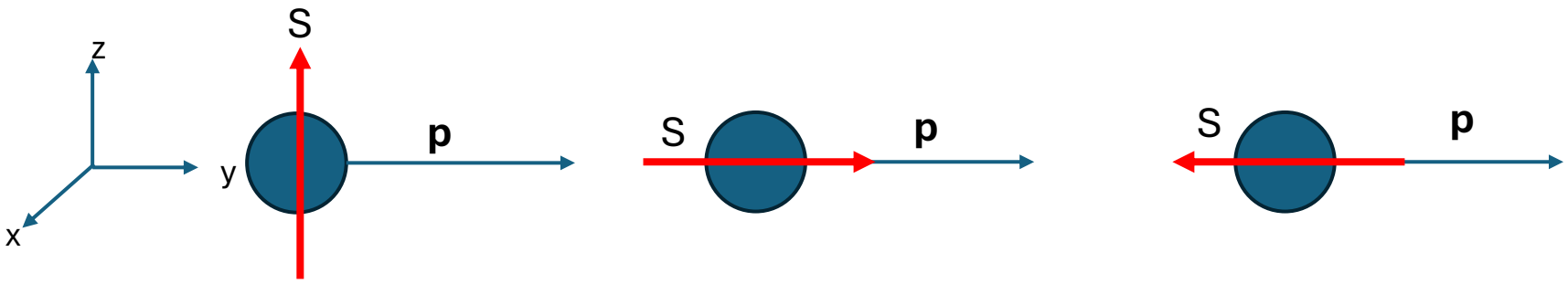
In the absence of a mass, left- and right-handed states are the eigenstates of the operator $\hat{\mathrm{h}}$, which is the helicity operator.

1. Chirality, helicity and spin

the helicity operator

$$h \equiv \hat{p} \cdot S = \frac{1}{2}\hat{p}_i \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix}$$

Helicity describes how the spin of a particle is aligned with its momentum.



h=0

h=1/2
Positive helicity: spin is parallel to momentum.

h=-1/2
negative helicity: spin is antiparallel to momentum.

For a massive particle, helicity is frame-dependent.
For a massless particle, helicity is not frame-dependent.

1. Chirality, helicity and spin

Property	Chirality	Helicity
Definition	Eigenvalue of γ^5	Projection to momentum to spin $\hat{\mathrm{p}} \cdot S$
Possible value	Left /Right	Positive /Negative
Operator	γ^5	$\hat{\mathrm{p}} \cdot \mathrm{S}$
Lorentz invariant?	Preserved under proper Lorentz transformations	Not generally for massive particles
Massless fermion	Corresponds to particle helicity	Correspond to particle chirality
Massive fermion	Both components generally present	Describes spin-momentum alignment.

Table 1. Summary of the difference

2. Solving the Dirac equation

For a free Dirac field, start with a positive-frequency plane wave.

$$\psi(x) = u(p)e^{-ip\cdot x}, \qquad p^0 = E_{\mathbf{p}} > 0$$

The spacetime dependence reduces the differential equation to algebra.

$$\left(i\gamma^\mu \partial_\mu - m\right)\psi = 0 \quad \Rightarrow \quad (\not{p} - m)u(p) = 0$$

$$g^{\mu\nu} = \mathrm{diag}(1,-1,-1,-1), \quad p^2 = m^2, \quad p\cdot x = Et - \mathbf{p}\cdot\mathbf{x}$$

We first solve this equation at rest, then boost the spinor.

2. Solving the Dirac equation

In the rest frame, the two chiral components must be equal.

$$p_0^\mu = (m, \mathbf{0}), \qquad m(\gamma^0 - I)u(p_0) = 0$$

$$\begin{pmatrix} -I & I \\ I & -I \end{pmatrix} \begin{pmatrix} u_L \\ u_R \end{pmatrix} = 0 \quad \Rightarrow \quad u_L = u_R$$

An arbitrary normalized two-component spinor labels the spin state.

$$u(p_0) = \sqrt{m} \begin{pmatrix} \xi \\ \xi \end{pmatrix}, \qquad \xi^\dagger \xi = 1$$

2. Solving the Dirac equation

A boost along z turns the rest momentum into a moving momentum.

$$\begin{pmatrix} E \\ p^3 \end{pmatrix} = \exp\left[\eta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right] \begin{pmatrix} m \\ 0 \end{pmatrix}$$

$$= \left[\cosh\eta \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \sinh\eta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right] \begin{pmatrix} m \\ 0 \end{pmatrix}$$

Finally,

$$E = m\cosh\eta, \qquad p^3 = m\sinh\eta, \qquad v = \tanh\eta$$

The rapidity connects the Lorentz boost to the energy and momentum.

$$E \pm p^3 = me^{\pm\eta} \quad \Rightarrow \quad \sqrt{m}\, e^{\pm\eta/2} = \sqrt{E \pm p^3}$$

2. Solving the Dirac equation

The same boost acts differently on the two chiral components.

$$u(p) = \exp\left[-\frac{1}{2}\eta\begin{pmatrix}\sigma^3 & 0\\ 0 & -\sigma^3\end{pmatrix}\right]\sqrt{m}\begin{pmatrix}\xi\\ \xi\end{pmatrix}$$

$$= \left[\cosh\left(\frac{1}{2}\eta\right)\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} - \sinh\left(\frac{1}{2}\eta\right)\begin{pmatrix}\sigma^3 & 0\\ 0 & -\sigma^3\end{pmatrix}\right]\sqrt{m}\begin{pmatrix}\xi\\ \xi\end{pmatrix}$$

$$= \begin{pmatrix}\left[\sqrt{E+p^3}\left(\frac{1-\sigma^3}{2}\right) + \sqrt{E-p^3}\left(\frac{1+\sigma^3}{2}\right)\right]\xi\\ \left[\sqrt{E+p^3}\left(\frac{1+\sigma^3}{2}\right) + \sqrt{E-p^3}\left(\frac{1-\sigma^3}{2}\right)\right]\xi\end{pmatrix}$$

Expand the matrix exponential using the Pauli-matrix identity.

$$(\sigma^3)^2 = I, \qquad e^{\pm\eta\sigma^3/2} = I\cosh\frac{\eta}{2} \pm \sigma^3\sinh\frac{\eta}{2}$$

2. Solving the Dirac equation

$$u(p) = \begin{pmatrix} \left[\sqrt{E+p^3}\left(\frac{1-\sigma^3}{2}\right) + \sqrt{E-p^3}\left(\frac{1+\sigma^3}{2}\right)\right]\xi \\ \left[\sqrt{E+p^3}\left(\frac{1+\sigma^3}{2}\right) + \sqrt{E-p^3}\left(\frac{1-\sigma^3}{2}\right)\right]\xi \end{pmatrix}$$

$$= \begin{pmatrix} \begin{pmatrix} \sqrt{E-p^3} & 0 \\ 0 & \sqrt{E+p^3} \end{pmatrix}\xi \\ \begin{pmatrix} \sqrt{E+p^3} & 0 \\ 0 & \sqrt{E-p^3} \end{pmatrix}\xi \end{pmatrix}$$

In our example(3-axis boost),

$$\sqrt{p\cdot\sigma} = \sqrt{EI - p^3\sigma^3} = \begin{pmatrix} \sqrt{E-p^3} & 0 \\ 0 & \sqrt{E+p^3} \end{pmatrix}$$

2. Solving the Dirac equation

Then an arbitrary momentum direction, the boosted solution is

$$u^s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma}\,\xi^s \\ \sqrt{p \cdot \overline{\sigma}}\,\xi^s \end{pmatrix}$$

$$p \cdot \sigma = EI - \mathbf{p} \cdot \boldsymbol{\sigma}, \quad p \cdot \overline{\sigma} = EI + \mathbf{p} \cdot \boldsymbol{\sigma}$$

Finally, we get general solution of the Dirac equation.

$$\psi(x) = \begin{pmatrix} \sqrt{p \cdot \sigma}\,\xi^s \\ \sqrt{p \cdot \overline{\sigma}}\,\xi^s \end{pmatrix} e^{-ip\cdot x}, \qquad p^0 = E_{\mathbf{p}} > 0, \qquad s = 1, 2, \qquad p^2 = m^2$$

2. Solving the Dirac equation

For eigenstate of σ^3

1. $\xi^s = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$u(p) = \begin{pmatrix} \sqrt{E - p^3}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \sqrt{E + p^3}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} \underset{\text{large boost}}{\longrightarrow} \sqrt{2E}\begin{pmatrix} 0 \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix}$$

2. $\xi^{s'} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$u(p) = \begin{pmatrix} \sqrt{E + p^3}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \sqrt{E - p^3}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} \underset{\text{large boost}}{\longrightarrow} \sqrt{2E}\begin{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 \end{pmatrix}$$

In the limit $\eta \to \infty$ the states degenerate into the two-component spinors of a massless particle, and the solutions are eigenstates of the helicity operator.

2. Solving the Dirac equation

The ordinary Hermitian norm depends on the energy of the frame.

$$u^\dagger u = \left(\xi^\dagger \sqrt{p\cdot\sigma},\ \xi^\dagger \sqrt{p\cdot\overline{\sigma}}\right)\cdot\begin{pmatrix}\sqrt{p\cdot\sigma}\,\xi\\ \sqrt{p\cdot\overline{\sigma}}\,\xi\end{pmatrix}$$

$$= 2E\,\xi^\dagger\xi$$

To make a Lorentz scalar

$$\overline{u}(p) = u^\dagger(p)\gamma^0$$

$$\overline{u}u = 2m\,\xi^\dagger\xi$$

The boost preserves the Dirac scalar normalization.

$$\xi^\dagger\xi = 1, \qquad \xi^1,\ \xi^2\text{:}\quad \begin{pmatrix}1\\0\end{pmatrix},\ \begin{pmatrix}0\\1\end{pmatrix}$$

The moving-frame norm follows from the rest-frame normalization.

2. Solving the Dirac equation

For the negative-frequency branch, reverse the sign in the plane wave.

$$\psi(x) = v(p)e^{+ip\cdot x}, \qquad p^0 = E_{\mathbf{p}} > 0$$

$$i\partial_\mu e^{+ip\cdot x} = -p_\mu e^{+ip\cdot x} \quad \Rightarrow \quad (\not{p} + m)v(p) = 0$$

At rest, the two chiral components have opposite signs.

$$(\gamma^0 + I)v(p_0) = 0 \quad \Rightarrow \quad v(p_0) = \sqrt{m}\begin{pmatrix} \eta \\ -\eta \end{pmatrix}$$

The energy label stays positive for both frequency branches.

2. Solving the Dirac equation

1. Normalize the negative-frequency spinors.

$$\overline{v}^r(p)v^s(p) = -2m\delta^{rs} \quad \text{or} \quad v^{r\dagger}(p)v^s(p) = +2E_{\mathbf{p}}\delta^{rs}$$

2. Dirac inner products separate the two solution branches.

$$\overline{u}^r(p)v^s(p) = \overline{v}^r(p)u^s(p) = 0$$

3. Hermitian overlaps need not vanish at equal momentum.

$$u^{r\dagger}(p)v^s(p) \neq 0, \qquad v^{r\dagger}(p)u^s(p) \neq 0$$

4. Reversing one three-momentum restores Hermitian orthogonality.

$$u^{r\dagger}(\mathbf{p})v^s(-\mathbf{p}) = v^{r\dagger}(-\mathbf{p})u^s(\mathbf{p}) = 0$$

2. Solving the Dirac equation

Sum the outer products over the two spin states.

$$\sum_{s=1,2} u^s(p)\overline{u}^s(p) = \sum_s \begin{pmatrix} \sqrt{p\cdot\sigma}\,\xi^s \\ \sqrt{p\cdot\overline{\sigma}}\,\xi^s \end{pmatrix} \left(\xi^{s\dagger}\sqrt{p\cdot\overline{\sigma}},\ \xi^{s\dagger}\sqrt{p\cdot\sigma} \right)$$

Use completeness of the two-component spin basis.

$$\sum_{s=1,2} \xi^s\,\xi^{s\dagger} = I$$

The spin labels disappear, leaving a matrix of momentum factors.

$$= \begin{pmatrix} \sqrt{p\cdot\sigma}\sqrt{p\cdot\overline{\sigma}} & \sqrt{p\cdot\sigma}\sqrt{p\cdot\sigma} \\ \sqrt{p\cdot\overline{\sigma}}\sqrt{p\cdot\overline{\sigma}} & \sqrt{p\cdot\overline{\sigma}}\sqrt{p\cdot\sigma} \end{pmatrix}$$

2. Solving the Dirac equation

The positive matrix square roots give the diagonal blocks.

$$\sqrt{p \cdot \sigma}\sqrt{p \cdot \overline{\sigma}} = mI$$

$$= \begin{pmatrix} m & p \cdot \sigma \\ p \cdot \overline{\sigma} & m \end{pmatrix}$$

In the Weyl basis, this is the covariant positive-frequency spin sum.

$$\sum_s u^s(p)\overline{u}^s(p) = \gamma \cdot p + m$$

The negative-frequency branch reverses the sign of the mass term.

$$\sum_s v^s(p)\overline{v}^s(p) = \gamma \cdot p - m$$

3. Charge conjugation, Parity and Time reversal

C, P, and T ask three different questions about a Dirac field.

Charge conjugation: what changes when a particle becomes its antiparticle?

Parity: what changes when all spatial coordinates are reversed?

Time reversal: what changes when the direction of time is reversed?

$$C,\ P\text{: unitary,} \qquad T\text{: antiunitary}$$

We focus on their physical meaning and the basic field transformations.

3. Charge conjugation, Parity and Time reversal

Parity reverses momentum, while spin behaves as an axial vector.

$$(t, \mathbf{x}) \mapsto (t, -\mathbf{x}), \qquad \mathbf{p} \mapsto -\mathbf{p}, \quad \mathbf{S} \mapsto \mathbf{S}$$

$$P\psi(t, \mathbf{x})P^{-1} = \gamma^0 \psi(t, -\mathbf{x})$$

In the Weyl basis, the parity matrix exchanges the two chiral blocks.

$$\gamma^0 \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix}, \qquad h \mapsto -h$$

For the free Dirac theory, parity leaves the dynamics unchanged.

3. Charge conjugation, Parity and Time reversal

Charge conjugation exchanges a particle with its antiparticle.

$$q \mapsto -q, \qquad \mathbf{p} \mapsto \mathbf{p}, \qquad \mathbf{S} \mapsto \mathbf{S}$$

The quantum Dirac field transforms as

$$C\psi(x)C^{-1} = -i\gamma^2\left[\psi^\dagger(x)\right]^T$$

C preserves the physical spin and helicity of the state.
The dagger in the field formula does not make C antiunitary.

$$CiC^{-1} = i$$

3. Charge conjugation, Parity and Time reversal

Time reversal reverses both momentum and angular momentum.

$$(t, \mathbf{x}) \mapsto (-t, \mathbf{x}), \quad \mathbf{p} \mapsto -\mathbf{p}, \quad \mathbf{S} \mapsto -\mathbf{S}$$

Time reversal preserves energy and changes the sign of i.

$$Te^{-iHt}T^{-1} = e^{+iHt}, \qquad THT^{-1} = H, \qquad TiT^{-1} = -i$$

$$T\psi(t, \mathbf{x})T^{-1} = \gamma^1\gamma^3\psi(-t, \mathbf{x})$$

This antiunitarity is the essential difference from C and P.

3. Charge conjugation, Parity and Time reversal

Helicity follows from the changes of momentum and spin.

$$h = \frac{\mathbf{S} \cdot \mathbf{p}}{|\mathbf{p}|}$$

C keeps both momentum and spin, so helicity is unchanged.

$$C\colon\ h \mapsto h, \qquad P\colon\ h \mapsto -h, \qquad T\colon\ h \mapsto h$$

Under T, the two minus signs cancel in the spin–momentum product.

$$\frac{(-\mathbf{S}) \cdot (-\mathbf{p})}{|\mathbf{p}|} = h$$

These statements refer to the physical particle or antiparticle state.

3. Charge conjugation, Parity and Time reversal

CPT combines charge conjugation, parity, and time reversal.

$$CPT: \quad x^\mu \mapsto -x^\mu, \quad q \mapsto -q, \quad \mathbf{p} \mapsto \mathbf{p}, \quad \mathbf{S} \mapsto -\mathbf{S}$$

The free Dirac theory respects C, P, and T separately.
Interactions can violate individual symmetries.

Local Lorentz-invariant QFT with Hermitian dynamics and the usual spin-statistics assumptions obeys CPT symmetry.

4. Summary

1. Chirality labels left/right Weyl components; mass couples them.

2. Helicity is spin along momentum; it depends on the frame for a massive particle.

3. Boosting rest spinors gives the u and v solutions, both with positive energy labels.

4. Normalization and spin sums give Lorentz-covariant identities.

5. C exchanges particle and antiparticle; P reverses helicity; T is antiunitary.

6. CPT holds under the standard assumptions of local, Lorentz-invariant QFT.